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Project Title:	Bioinspired Zeolite Assemblies as Artificial Photosynthetic Systems		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
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Zip Code:	43210-1173	**Congressional District:**	15
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	George, Michael (University of Alabama in Huntsville)		
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	Our goal is to assemble a supramolecular structure around a central zeolite membrane with the following features. · Energy transfer from extrazeolitic molecules to sensitizer molecules on the surface of the zeolite membrane · Charge (electron) transfer from the photoexcited sensitizer molecule to acceptor molecules within the zeolite membrane · Charge transport through the membrane by electron hopping · Catalytic oxidation and reduction of H2O to oxygen and hydrogen on two opposite sides of the membrane and regeneration of the donor and acceptor		

Task Description:	· Zeolite pore engineering to ensure entrapment of photoactive molecules, but still allow for proton migration to maintain charge balance
	
	The task is divided into four subcategories, dealing with zeolitic energy transfer, electron transfer processes on the surface of the zeolite, intrazeolitic charge propagation and zeolite-based nanostructured catalysts. Choice of molecules include polypyridyl complexes of Ru and Os as sensitizers and electron donors and bipyridinium-based systems as electron acceptors and metal oxides as catalysts. Photosynthesis, the conversion of sunlight to useful chemical energy is the basis for life. Important concepts that have emerged from studies on natural photosynthesis include the structure of the photoactive molecules, novel spatial arrangement, directed electron transfer and catalysis. Our objective is to design an artificial photosynthetic assembly with the goal of photolytic splitting of water into hydrogen and oxygen. We propose to use microporous crystalline, aluminosilicate zeolites as the host system. The challenge in design of complex, multicomponent functional assemblies from nanoscale building blocks is the optimization and assembly of the various components. Since the ordered array of channels and cages exist in nanocrystallites as well as in larger zeolite membranes, a convenient template for self assembly over large length scales becomes possible. Choice of molecules in this study include polypyridyl complexes of Ru and Os as sensitizers and electron donors and bipyridinium-based systems as electron acceptors. The nanometer size cages and channels of aluminosilicate zeolites will anchor the photochemical moieties as well as the catalysts necessary for reaction with charge separated redox species. The specific objectives of the research program includes:
	
	$ energy transfer within nanocrystalline (~20-100nm) zeolite crystallites, as well as energy transfer into and out of a zeolite
	
	$ electron transfer dynamics from surface-bound sensitizers into acceptor molecules in zeolites
	
	$ synthesis of zeolitic membranes from nanocrystalline zeolites and photochemical charge propagation within zeolite membranes
	
	$ use of nanometer sized zeolite cages for synthesis of metal oxide catalysts for multi-electron processes
	
	· assembly of the above components into a multifunctional unit capable of light-driven conversion of water to H2 and O2. Long term space exploration and habitation necessitate that systems be developed for generation of molecular building blocks that can be assembled to form complex macroscopic structures, such as fuel, polymers, drugs, etc. Ideally, the process should begin with the simplest of molecules that are readily available and use a plentiful energy source. The best example in nature of such a process is photosynthesis, which, in plants, involves the solar energy-driven catalytic reaction of making carbohydrates from water and carbon dioxide. This is the basis on which life is sustained on earth. The conversion of H2O and CO2 to O2 and carbohydrates is a complex redox process, involving a large array of molecules and a complex membrane-based architecture. In principle, many solar energy-based transformations are possible, in which simple molecules like H2O, CO2, N2 can be converted to molecular building blocks such as H2, CH4, NH3, which can form the basis for manufacture of important chemicals necessary for life.
	
	Solar radiation can drive these uphill reactions. The simplicity of these reactions masks the complexity of the architecture necessary to accomplish the goal. However, natural photosynthesis provides significant clues for assembly of "artificial" photosynthetic systems, and in this proposal, we propose a zeolite-based architecture, that at a primitive level mimics natural photosynthesis. For our model redox reactions, we have chosen the splitting of H2O into H2 and O2. The principles we develop in this program should be applicable to other photoredox systems, especially the architectural aspects related to assembly of multifunctioning, multicomponent units. The limitations of weight that can be carried on long term missions as well as space habitation require that materials taken from earth can serve several functions. Zeolites, because of their applications in adsorption, ion-exchange, separations and catalysis can play that role. In addition, photosensitized reactions for both environmental pollution control (air and water purification) as well as synthesis of fine chemicals have been reported with zeolites, and is currently satisfying aspects of “sustainable/green chemistry” on earth .

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful artificial photosynthesis systems that can manufacture fuels and chemicals using sunlight and simple chemicals such as water and carbon dioxide as feed will revolutionize our energy dependence. These technologies are examples of "disruptive technologies", since they set up a new paradigm of energy utilization.
Task Progress:	Development of a zeolite membrane for charge transfer: Photochemically generated long-lived charge separation is the key step in processes that aim for conversion of solar into chemical energy. In this study, we focus on a Ru polypyridyl complex [(bpy)2 Ru(II)L , bpy = bipyridine, L = 1, 2-bis [4-(4’ – 2,2’ – bipyridyl) ethene] encapsulated on the surface of a pin-hole free zeolite membrane by quaternization of L and surrounded with intrazeolitic bipyridinium ions (N,N’-trimethyl-2,2’-bipyridinium ion, 3-DQ2+). Visible-light irradiation of the Ru complex side of the membrane in the presence of a sacrificial electron donor led to formation of PVS? • on the other side. Pore-blocking disilazane-based chemistry allows for Na+ to migrate through the membrane to maintain charge balance, while keeping the 3-DQ2+ entrapped in the zeolite. These results provide encouragement that the zeolite-membrane based architecture has the necessary features for not only incorporating molecular assemblies with long-lived charge separation, but also for ready exploitation of the spatially separated charges to store visible light energy in chemical species. Surface modifications of zeolites: The surface hydroxyl groups of zeolite Y were reacted with 1,1,3,3-tetramethyldisilazane, HN(SiHMe2)2 under ambient conditions and the grafting of siloxy functionality on the zeolite was confirmed by infrared, NMR spectroscopy and elemental analysis. Paraquat (methyl viologen) was ion-exchanged into the zeolite, followed by treatment with the disilazane reagent. Surface modification of paraquat-loaded zeolites encapsulates the guest molecules in the zeolite cages and release of paraquat by ion-exchange with sodium ions was studied. The total amount of paraquat released was dependent on the concentration of Na+ in solution, and was similar for the derivatized and underivatized samples. In the absence of surface modification, equilibration occurred within 20 minutes, whereas with surface modification, the equilibration time was extended to 7 days . These kinetics are reflected in the effective diffusion coefficients (D) of paraquat, with D = 1.2 × 10-15 cm2 s-1 for derivatized zeolite Y and D = 0.2-1.1 × 10-7 cm2 s-1 for the underivatized sample. Paraquat was chosen as the guest molecule, since it is widely used as an herbicide and its controlled release is of interest in agricultural applications.

Bibliography Type:	Description: (Last Updated: 03/10/2008)
Articles in Peer-reviewed Journals	Kim Y, Das A, Zhang H, Dutta PK. "Zeolite membrane-based artificial photosynthetic assembly for long-lived charge separation." J Phys Chem B. 2005 Apr 21;109(15):6929-32. PMID: 16851782 , Apr-2005
Presentation	Prabir Dutta "Biomimetic Photosynthetic Assemblies " National Space and Missile Materials Symposium Jun-2005
Presentation	Prabir Dutta "Artificial Photosynthetic Assemblies for Water Photsplitting" American Chemical Society National Meeting, San Diego Mar-2005
Presentation	Prabir Dutta "Zeolites as Artificial Photosynthetic Systems" NASA Marshall Mar-2005